

Electrochemistry Summary

Definition: electrochemistry is the process of using electricity to break down or decompose a compound into its separate components.

must be soluble
 refer to solubility table of salts
 Ionic compound / Covalent with ionic character
 e.g. HCl / H_2SO_4
 which ionize to prod. respective ions.

electrolysis → important → extraction of useful pure elements from molten / aq compounds
 e.g. of useful elements.
 Very reactive metals
 like group 1 metals and non-metals like chlorine

Definition: Electrolytic cell is a device that converts electrical energy to chemical energy.

Electrolytic cell

1. Batteries
2. Positive electrode - anode
 ↳ the terminal of battery
3. Negative electrode - cathode
 ↳ a -ve terminal of battery
4. connecting wires
5. Electrolyte

uses electricity to drive non-spontaneous chemical reaction (redox reaction)
 doesn't occur on its own without external energy input
 Energy conversion: Electrical energy → chemical energy
 requires continuous energy supply to proceed (heat, electricity, light)

Electrodes

Unreactive (Inert)

- Graphite
- Platinum

Inert anode: Remain chemically unchanged (does not undergo oxidation/reduction) when current flows thru.

- Anode does not interfere with type of ions discharged
- Does not react with electrolyte / products

Reactive

- Any metal above Platinum in reactivity series

Reactive anode: will undergo chemical reaction (oxidation) when current flows through.

used in purification
 No ions discharged at anode ever.

Interferes with type of ions discharged

Important:

Electrons flow from anode to cathode.

- negative terminal to positive terminal of battery.

Similarity: Cathode

→ remains chemically unchanged (no oxidation/reduction) when current flows thru.

→ Does not interfere with discharge of ions

→ Does not react with electrolyte / products of electrolysis.

Electrolytes

- Compounds which are good conductors when molten / aqueous
- Electrolytes → electrolytic conductors → decompose → form new substances when they conduct electricity
- Electricity → conducted → due to presence of free / -ve ions → move freely across molten / aq electrolyte → and act as charge carriers.
- usually ionic compounds / water soluble covalent compounds.

Strong electrolytes

- Compounds that ionize completely to form high concentration of mobile ions
- Greater concentration → greater capacity to carry charges / conduct electricity.

Strong Acids

Strong Alkalies

Non-electrolytes

- Do not ionize!

Weak electrolytes

- Ionize partially → low concentration of mobile ions.
- lower concentration of mobile ions → lower capacity to carry charges / conduct electricity.

Weak Acids

Weak Alkalies

Basic Step for Electrolytic cell with Inert electrodes

1. Switch in external circuit closed → electrolysis occurs → battery provides energy required to drive process

- External circuit — electrons flow from the -ve terminal to the terminal. → From anode to cathode. thru external circuit

At positive electrode (anode)

- attracts anions. Anions thus migrate to anode.
- Since anode is inert → anion undergo oxidation → lose electrons
- Half equation in the form:
e.g. $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

Does Remains Chemically unchanged
Does not interfere with type of ions discharged.

At negative anode (cathode)

- attracts cations. Thus cations migrate to cathode.
- Since cathode is inert → cations undergo reduction → gain electrons.
- Half eqn in the form:
e.g. $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$

Note: ions do not flow thru electrolyte but instead wire as wire has lower resistance than electrolyte.

- Electrolyte conducts electricity to form the closed circuit via ions.

Electrolysis of Molten Ionic compounds using Inert Electrodes

Electrolyte must be heated strongly throughout process:

• Only when electrolyte is molten/solution can ions move freely throughout substance and migrate to opp. charged electrodes + and electrolysis can take place.

• Properties of ionic compounds → high melting + Boiling point.

↳ P.T.P. → NaCl → solid state

Properties of ionic $\star$

↳ Large amt of heat has to be provided → Overcome strong EFA which hold Na^+ and Cl^- ions in fixed positions in giant ionic crystal lattice structure to melt substance.

once substance has melted + it has to be heated continuously → ensure temp. remains above melting point for electrolysis to take place continuously.

If heating stops → temp. of NaCl → will eventually decrease + until NaCl solidifies and electrolysis will no longer take place → ions can no longer move freely and be attracted to opp. charged electrodes.

Answer Technique

• Correct way to describe reaction at electrodes

e.g. →

- chloride ions discharged at anode as they undergo oxidation + losing electrons to form chlorine molecules
- Sodium ions discharged at cathode as they undergo reduction to form sodium atoms.

Incorrect way:

e.g. Sodium discharged at cathode and chlorine discharged at anode.

Predicting Products of Electrolysis

1. Identify cations + anions

For anions:

i) oxygen formed if solution:

↳ has sulfates/nitrates

↳ or is too dilute

ii) halogens formed if → halides present in high concentration

For cations:

↳ Refer to reactivity series $\star$

↳ ions of metals above hydrogen → will not be discharged → ions remain in solution
↳ H_2 gas discharged instead

• ions of metals below hydrogen → discharged → metal deposited on electrode.

Identify cation + anion remaining in solution (products formed in solution)

Solution may cause:

• Change in pH value (if OH^- or H^+ discharged → resulting in concentration change of ion)

• Increase in concentration (if both OH^- and H^+ discharged)

• Change in colour intensity
e.g. Cu^{2+} discharged to form Cu → blue fades to colourless

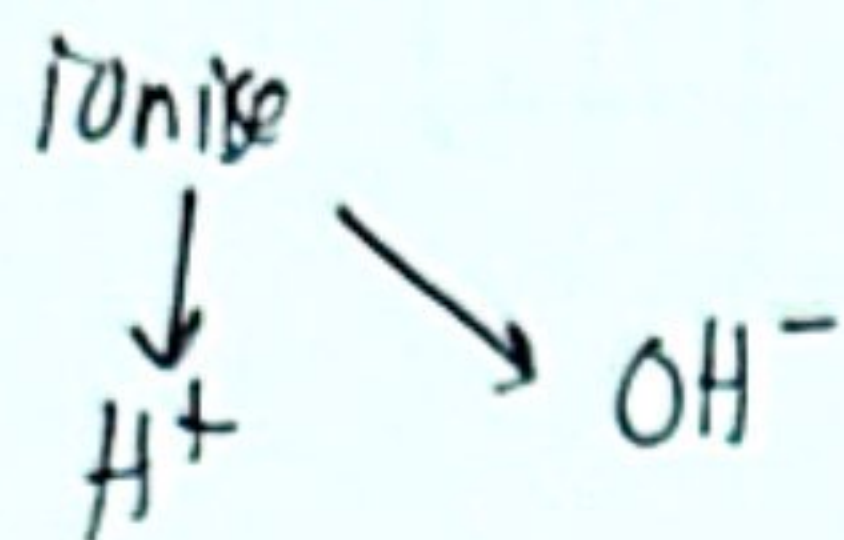
Eg. Sodium:

- Tiny globules of molten sodium float to surface of molten sodium chloride
- sodium burns in air with flashes of yellow light.

Electrolysis of aqueous solution using Inert electrode

- When 2 type of cation/anion → only 1 cation and 1 anion selectively discharged ^{Or preferentially}
- Note: Both cations / Both anions are attracted to their respective electrode

H₂O can also



Rules of selective discharge

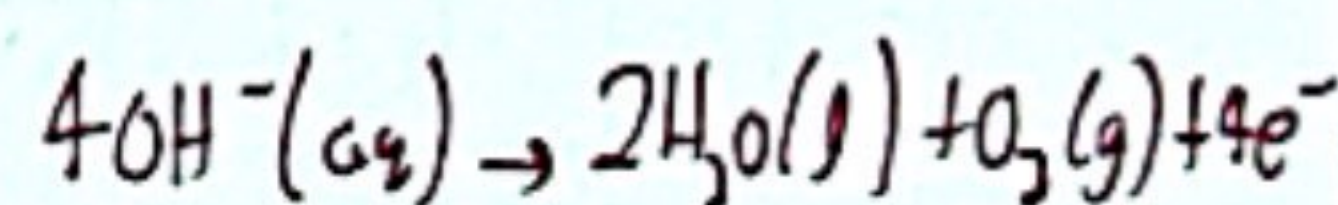
Cations

- metal ions of element lower in reactivity series discharged in preference to other cations.

less energy is required!
Cations of less reactive metal gain electrons more readily
↓
greater tendency to undergo reduction

Above hydrogen → not discharged
Below hydrogen → discharged

Standard eqn for hydroxide discharged:



Match electrons of cation to match this
4e⁻
↳ as electrons discharged on both sides must be same.

Anions

- Dilute aq solution / concentrated solutions that do not contain halide.
- Solutions that contain sulfate and nitrate, regardless of concentration
↳ OH⁻ preferentially / selectively discharged in preference to other anions → less energy required

↓
OH⁻ ions → lose electrons more readily → greater tendency to be oxidised.

- concentrated solutions that contain halide ions
↳ halide selected for discharge in preference to OH⁻
↳ bc higher concentration of halide ions is higher than concentration of OH⁻ ions

- dilute solutions → contain halide
↳ hydroxide selectively discharged in preference to halide ions
↳ based on ease of discharge of ions.

Polyatomic anions → Sulfate / Nitrate

↳ not discharged → remain in solution → remain in solution

Not affected by concentration, higher or lower.

↳ more energy required → less easily oxidised.

Electrolysis of Dilute Sodium Chloride
→ results in decomp. of water → into components.

→ water removed from NaCl solution → concentration of NaCl increases.

Note: This method cannot be used for reactive metals as they won't be discharged!
↳ we molten instead!

Imp! If both OH⁻ and H⁺ removed → not change in acidity / alkalinity

Ions selectively discharged, not substance!!

Explain why volume of H_2 produced at cathode is 2x volume of O_2 prod. at anode.

In this electrolytic cell, 4 moles of electrons transferred from anode to cathode which results in formation of 2 mol of H_2 gas at cathode but only 1 mol of O_2 gas at anode.

According to Avogadro's law 1 mol of gas occupies $24 dm^3$ at r.t.p. thus for the reaction, mol ratio of hydrogen gas to oxygen gas prod. is same as volume ratio of gases which is 2:1. Thus volume of hydrogen gas produced during electrolysis is 2x vol. of oxygen.

During electrolysis observed that mass of anode decreases and white ppt forms.

As E progresses $\rightarrow$ use of electricity causes electrodes to heat up over time. At higher temps O_2 gas produced at anode will react with carbon anode to produce CO_2 gas resulting in loss of mass of anode.

CO_2 produced reacts with $CaOH$ (lime water electrolyte) $\rightarrow$ form white ppt of $CaCO_3$ insoluble

Electrolysis of Ag solution using Reactive electrodes

$\rightarrow$ Electroplating!
 $\rightarrow$ Purification of metals!

Electrolyte made of any metal above platinum in reactivity series.

Phrase:

Copper anode, which is reactive anode electrode $\rightarrow$ loses electrons more readily than anions $\rightarrow$ will undergo oxidation instead of anions.

Greater tendency to undergo oxidation!

concept of selective discharge of cations still applies $\rightarrow$ cations reduced at cathode

concept of selective discharge of anions DOES NOT APPLY $\rightarrow$ reactive anode undergoes oxidation instead of anions

No anion discharged! $\rightarrow$ Remain in solution

Observations

Copper anode slowly becomes smaller (loses mass)

$\rightarrow$ Dissolves gradually in electrolyte

$\rightarrow$ Cu anode loses electrons more readily than anions thus Cu anode undergoes oxidation in place of anions

$\rightarrow$ $Cu^{2+}(aq)$ ions enter aq solution of $XXXX$

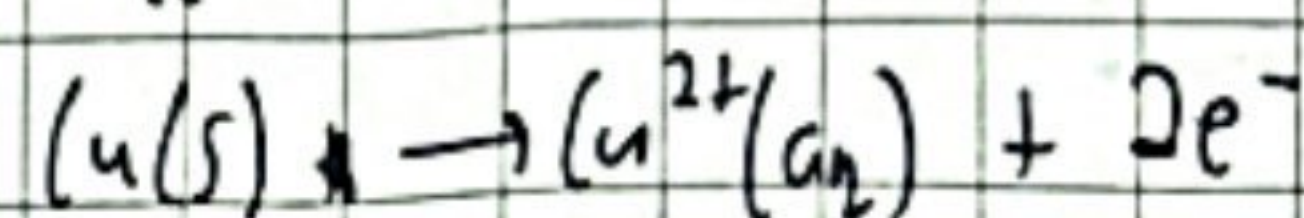
Copper cathode becomes larger/gain mass

$\rightarrow$ Cu^{2+} gain electrons more readily than H^+ $\rightarrow$ Cu^{2+} selectively discharged at cathode.

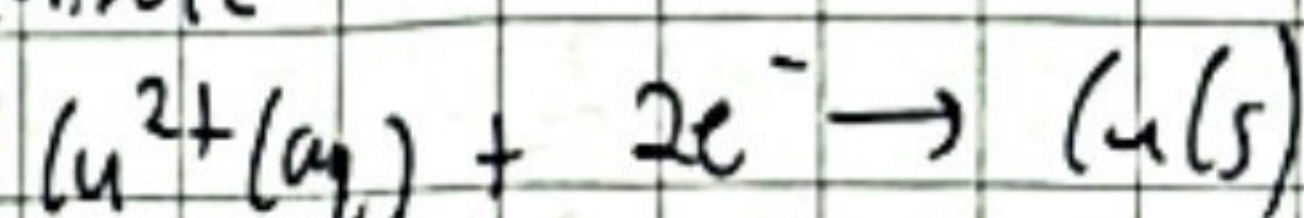
$\rightarrow$ Cu^{2+} reduces $\rightarrow$ form Cu which coats pure copper cathode/deposited on pure copper cathode $\rightarrow$ coating anode in size.

Half eqn example:

Anode:



Cathode



Products formed

Anode:

Cu^{2+} ion

Cathode:

Copper metal deposited

Blue colour of $Cu(NO_3)_2$ remains unchanged

$\rightarrow$ For every Cu^{2+} ion discharged at cathode to form Cu, there is 1 Cu^{2+} ion formed in electrolyte as 1 atom of Cu oxidises to form Cu^{2+} that enters electrolyte

$\rightarrow$ Thus concn of Cu^{2+} ions which is responsible for blue colour $\rightarrow$ remains unchanged. Thus blue colour unchanged.

Electrolysis of Ag Solutions using Reactive electrodes

Purification of Metals (Electrolytic refining of metals)

Application: purification of copper → make electrical wiring (Requires copper to be very pure)

Components:

- Battery
- Connecting wires
- Positive Electrode (anode): Impure metal ★
- Negative Electrode (cathode): thin sheet of pure form of impure metal ★
- Electrolyte: Ag solution containing cations of metal to be purified (solution of soluble salt of metal)

Eg: purification of silver:
Soluble silver salt solution

Must include term solution / aqueous for electrolyte as
solid cannot conduct electricity process won't start

Observation: Identical to previous page of Reactive electrodes.

Extra: Anode slime / Anode sludge is formed below anode

→ Metal impurities such as silver and platinum → did not undergo oxidation.

→ Impurities form fall to bottom of electrolytic cell, form 'anode slime' below anode

Electroplating

- Form thin protective coating of metal on surface of another which is likely to corrode
- Make metal more attractive / protect from corrosion. Plating metals → Corrosion resistant + shiny → they are lower reactivity series

Components:

- Battery
- Connecting wires
- Positive electrode (anode): plating metal
- Negative electrode (cathode): object to be plated
- Electrolyte: Ag solution containing cations of plating metal.

eg: silver plating
in Ag solution
solution of soluble / Ag / solution

Observations

- Silver anode slowly becomes smaller / reduce mass
- Disturbance gradually re-electrolyte
- Ag anode - loses electrons more readily than anions → thus Ag anode undergoes oxidation in place of anions
- Silver ions enter aq. solution of silver nitrate.

- metal spoon coated with layer of shiny, silver solid

- Ag^+ gains electron more readily than H^+ → thus Ag^+ is selectively discharged at cathode
- Ag^+ → reduced → form Ag → which coats spoon

Note: Usually same colour remains during electroplating → (concentration of cations in solution remains same)

To complete circuit → object to be electroplated → should be good conductor (CWE)
Poor (CWE) → need to be coated with thin layer of graphite before it can be electroplated.

Zinc → dustbin, bucket

Chromium → water tap, car bumper, bicycle handle bar

Tin → tin can

Silver → cutlery, trophies

Gold → jewellery, watches

Industrial Applications of Electrolysis

1. Electrolysis of molten compounds using inert electrodes → extract reactive metals from ores

Simple cells (Electric, Galvanic, Voltaic cells)

• Device uses chemical ~~reactions~~ reactions to generate electricity

Energy conversion:

Chemical energy $\rightarrow$ Electrical energy

Components:

- bulb / voltmeter

- negative electrode (anode)

$\rightarrow$ similar to -ve terminal of battery

$\rightarrow$ More reactive metal electrode

- positive electrode (cathode)

$\rightarrow$ similar to +ve terminal of battery

$\rightarrow$ less reactive metal electrode

- connecting wires (connect bulb/voltmeter to anode and cathode)

- electrolyte (3 Types)

$\rightarrow$ Aq. solution of compounds of group 1 metals (salt solution)

$\rightarrow$ Aq. solution of compound of less reactive metal which is used to make cathode (salt solution of less reactive metal)

$\rightarrow$ dilute acids

★ Greater diff. in reactivities of 2 metal electrodes

$\rightarrow$ the greater the p.d. (voltage) prod.

$\rightarrow$ Greater amt of electricity generated $\rightarrow$ Brighter bulb

★ why is anode more reactive metal electrode (-ve electrode)

$\rightarrow$ More reactive metal $\rightarrow$ loses electrons more readily $\rightarrow$ to form free ions $\rightarrow$ than less reactive metal

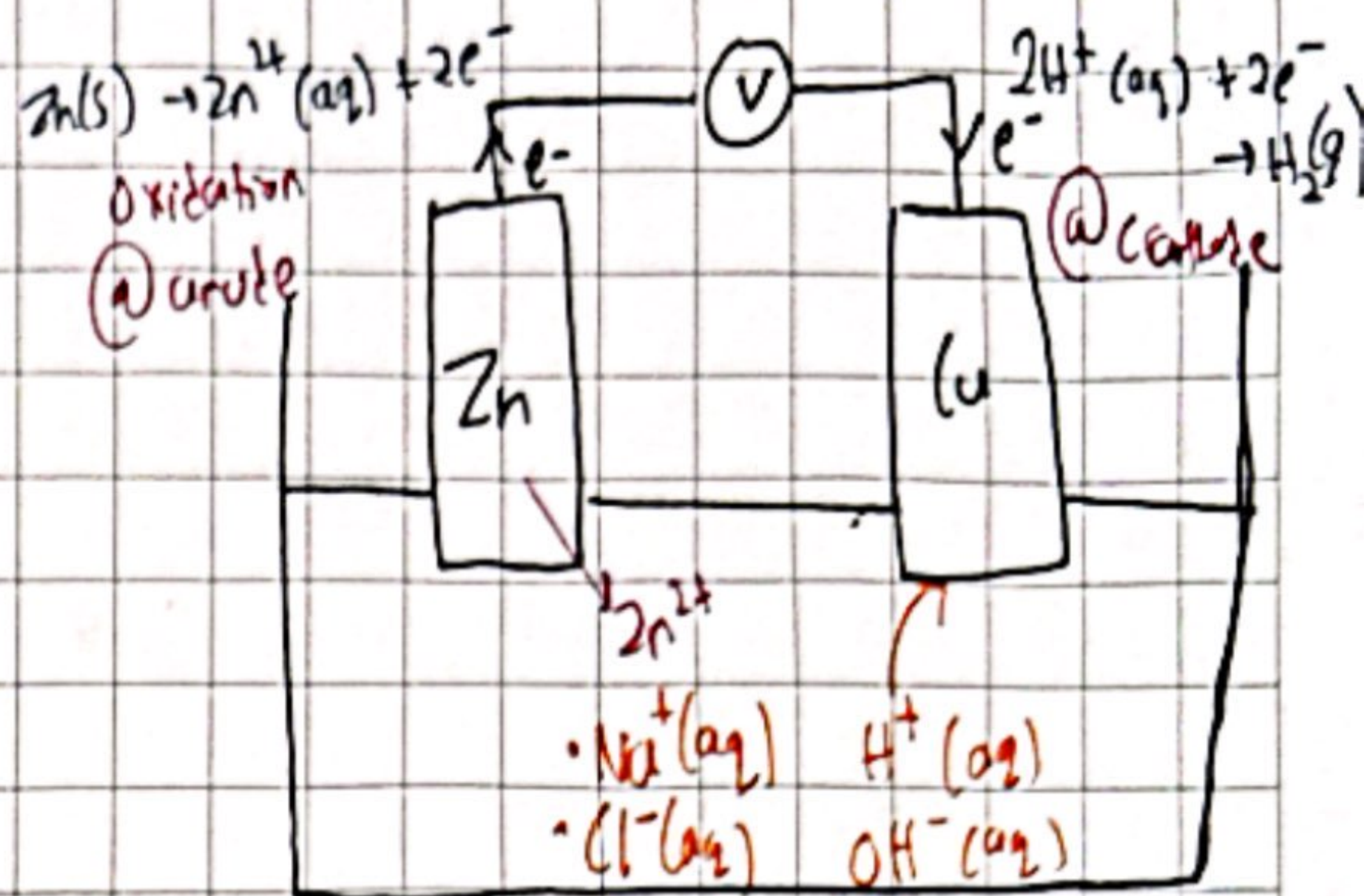
$\rightarrow$ Thus more reactive metal electrode $\rightarrow$ anode $\rightarrow$ as oxidation occurs at anode

$\rightarrow$ -ve charged electrons produced at anode $\rightarrow$ thus it is -ve electrode.

• Battery is made up of simple cells.

• chemical reactions occurring in simple cell produce electricity $\rightarrow$ which is used to drive non-spontaneous redox reactions in electrolyte (a.s.).

$\rightarrow$ Thus battery not part of set-up



• unlike electrolytic cell $\rightarrow$ negative electrode in simple cell is anode, while positive electrode is cathode

• like electrolytic cell $\rightarrow$ reduction takes place at cathode and oxidation takes place at anode

★ why is cathode less reactive metal electrode (+ve electrode)

$\rightarrow$ negatively charged electrons produced $\rightarrow$ will flow from negative to positive.
 • Since electrons produced at anode $\rightarrow$ anode is negative electrode while cathode is positive electrode

$\rightarrow$ electrons flow thru external circuit $\rightarrow$ from more reactive metal electrode to less reactive metal electrode $\rightarrow$ where it will be involved in reduction of cation.

$\rightarrow$ Thus less reactive metal electrode $\rightarrow$ as cathode or cathode is site for reduction

Concept of selective discharge
of cations still applies

↳ cation reduced at
cathode

Concept of selective discharge
of anions DO NOT APPLY

↳ anode - which is reactive electrode
undergoes oxidation instead
of anions.

Comparison of 3 simple cells which have
diff. electrolytes

• Potassium chlorate solution

★

• Ag Silver Nitrate

• Dilute Sulfuric Acid

★

cells use Magnesium and Silver
electrodes

Advantage

Portable source of electrical
energy → can be used in places
where there is no connection
to power supply from Mains.

Disadvantage
Expensive